Department of Computer Science and Engineering
Shiv Nadar University Chennai
July-November 2026 Semester
M.Tech. Course on Deep Learning
Lab Assignment 1

Date: 4th August, 2026

1. **Consider the 4-dimensional vectors, $\mathbf{x}, \mathbf{y}, \mathbf{z}$ given below. Compute the Euclidean distance between (i) x and y, (ii) x and z.**

   $x_1 = 3 \quad x_2 = -4 \quad x_3 = 6 \quad x_4 = -1$

   $y_1 = -2 \quad y_2 = 5 \quad y_3 = 8 \quad y_4 = -3$

   $z_1 = 5 \quad z_2 = 3 \quad z_3 = -1 \quad z_4 = -4$

2. **Consider the vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$ in Question (1). Compute the cosine similarity between (i) x and y, (ii) x and z.**

3. **Compute and plot each of the following activation functions:**
   (a) **Logistic sigmoid with $\beta = 0.5, 1.0, 2.0$**
   (b) **Hyperbolic tangent with $\beta = 0.5, 1.0, 2.0$**
   (c) **ReLU**
   (d) **Smooth ReLU (SoftPlus)**
   (e) **Leaky ReLU with $\delta = 0.5, 1.0, 2.0$**
   (f) **Swish function with $\beta = 0.5, 1.0, 2.0$**

4. **For each of the activation functions in Question (3), plot the derivative of the action function with the activation value.**

5. **Consider a multilayer feedforward neural network with a single hidden layer, to be used for a classification task. The number of nodes is 2 in the input layer, 4 in the hidden layer, and 3 in the output layer. The activation function used in the hidden layer nodes is the tanh function with $\beta = 1.0$. The output layer nodes use the logistic sigmoid with $\beta = 1.0$. The sum of squared errors is used as the loss function. An input vector x belonging to Class 2, and the weight matrices are as given below. The pattern mode of training is used. The learning rate $\eta = 0.9$.**

Input vector: $x_1 = 2.5 \quad x_2 = -4.3$

Weights of connections between the input layer and the hidden layer:

$w_{01}^h = 0.2 \quad w_{11}^h = -1.6 \quad w_{21}^h = 0.3$

$w_{02}^h = -0.4 \quad w_{12}^h = 0.5 \quad w_{22}^h = -0.7$

$w_{03}^h = 0.6 \quad w_{13}^h = 1.3 \quad w_{23}^h = 0.8$

$w_{04}^h = -0.9 \quad w_{14}^h = -0.3 \quad w_{24}^h = 1.4$

Weights of connections between the hidden layer and the output layer:

$w_{01}^o = 0.1 \quad w_{11}^o = 0.2 \quad w_{21}^o = -0.5 \quad w_{31}^o = 2.6 \quad w_{41}^o = -1.6$

$w_{02}^o = -0.6 \quad w_{12}^o = 0.7 \quad w_{22}^o = 1.3 \quad w_{32}^o = -3.2 \quad w_{42}^o = -2.5$

$w_{03}^o = 0.7 \quad w_{13}^o = -1.7 \quad w_{23}^o = 2.5 \quad w_{33}^o = 0.2 \quad w_{43}^o = 3.2$

(a) **Perform the forward pass of computation to compute the values of the outputs of the nodes in the hidden layer and the output layer.**

(b) **Compute the local gradients of the nodes in the output layer and the hidden layer.**

(c) **Determine the change in each of the parameters of the model.**

6. **Consider a training dataset of 10 examples represented by 2-dimensional feature vectors. The values of each of the features for the examples are given below.**

x_1 : $-2, \quad 5, \quad -3, \quad 7, \quad 4, \quad -1, \quad 8, \quad -6, \quad 0, \quad 1$

x_2 : $12, \quad 15, \quad -13, \quad -6, \quad 4, \quad -7, \quad 13, \quad -2, \quad 10, \quad -5$

Compute values of the normalized features obtained using (i) Min-Max Normalization and (ii) Mean-Variance Normalization.